Alltech Solutions



CAN BUS, EXPLAINED

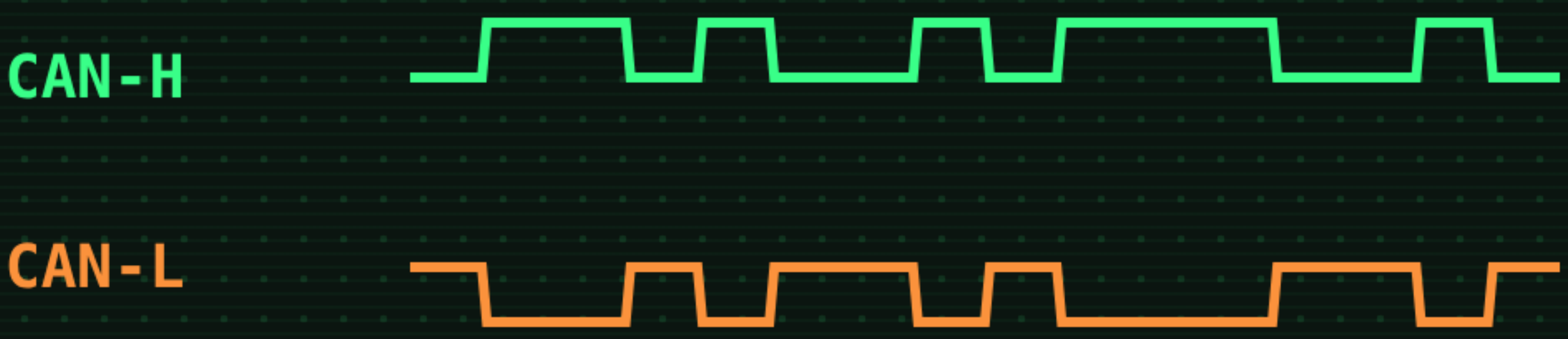
# What does a car's CAN bus actually say?

Every module in a modern car talks over two wires. Here is one real message, read off a scope.

Swipe ▶

# Two wires, mirrored

CAN-H and CAN-L both rest at 2.5 V. For each “0” bit, CAN-H rises to 3.5 V and CAN-L falls to 1.5 V.




The ECU reads the difference, so noise that hits both wires cancels out.

# One message, five parts



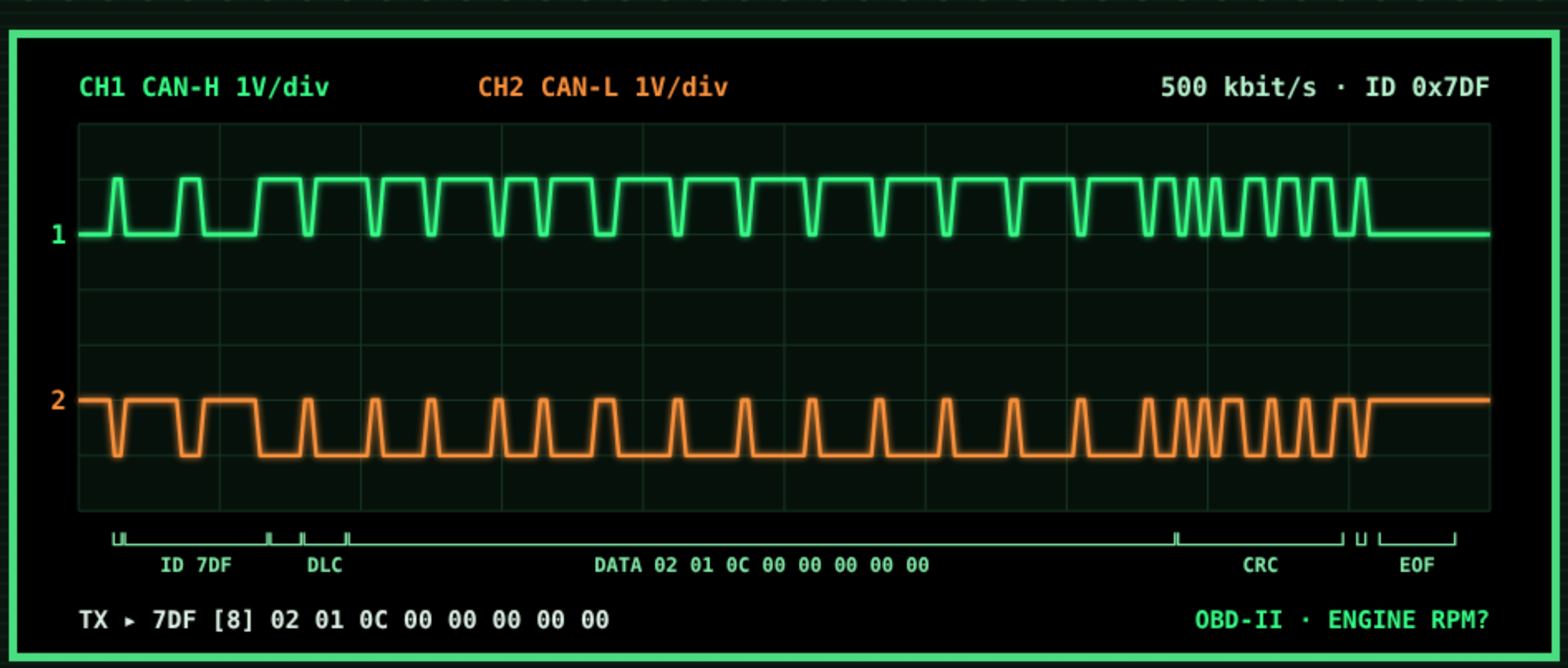

**ID** says what the message is about, and who wins if two modules talk at once: the lower ID goes first.

**DATA** carries up to 8 bytes. **CRC** lets every module check nothing got scrambled.

# A scan tool asking for engine RPM

ID 7DF · 8 bytes

| 02 | 01 | 0C | 00 | 00 | 00 | 00 | 00 |
|---|---|---|---|---|---|---|---|
| 2 bytes follow | show current data | engine RPM | padding | | | | |



# When modules stop talking, the car stops too

A dead ECU, a mismatched immobiliser or a replacement module that hasn't been programmed all look like "won't start".

Alltech Solutions programs, clones and virginises ECUs in Lagos, and writes the procedures up in a free database.

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